

Replicating ATE and ATET Estimates from *High-Dimensional Metrics in R*

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GitHub Link: <https://github.com/Patrick137469/hdm-401k-replication.git>

1 Introduction

This study replicates **High-Dimensional Metrics in R** written by **Victor Chernozhukov, Christian Hansen, and Martin Spindler** and published in **The R Journal** in 2016 (Chernozhukov et al., 2016). The paper studies high-dimensional econometric methods for prediction and causal inference when the researcher has many possible control variables.

The contribution of this paper is that it gets around the following challenge: **standard regression methods can become unreliable when there are many possible controls relative to the sample size, especially when the researcher wants valid inference after variable selection.** It does this by using rigorous **LASSO** and **Post-LASSO** methods with orthogonalized estimating equations for valid causal inference in high-dimensional settings.

This replication focuses on **Section 6.3, the 401(k) plan participation application** from the original paper. Specifically, the paper reports that **the estimated average treatment effect (ATE) is 10,180.09 with a standard error of 1,930.68, and the estimated average treatment effect on the treated (ATET) is 12,628.46 with a standard error of 2,944.43.**

2 Data

The data are **the pension dataset included in the hdm R package** and contain information on **401(k) participation, 401(k) eligibility, household wealth, income, age**

categories, family size, education, marital status, home ownership, IRA participation, and other household characteristics. The sample includes 9,915 observations.

The data were originally collected from the **1991 Survey of Income and Program Participation (SIPP)**. The key variables used in the replication are **total wealth, 401(k) participation, income category controls, age category controls, family size, education indicators, marital status, two-earner household status, defined benefit pension status, IRA participation, and home ownership**.

The original code is written in **R** and this replication reproduces the analysis in **R**.

3 Methods

The primary method is **high-dimensional treatment effect estimation using rigorous LASSO/Post-LASSO**, which does **variable selection among many possible controls while still allowing valid inference for the treatment effect**. The treatment effect methods are related to post-selection inference methods developed by Belloni et al. (2014).

The method predicts/identifies **the effect of 401(k) participation or eligibility on total wealth** by exploiting **observed household-level covariates and LASSO-based adjustment for confounding factors**.

The estimating equations used in the paper are shown below:

$$ATE = E[Y(1) - Y(0)] \tag{1}$$

$$ATET = E[Y(1) - Y(0) \mid D = 1] \tag{2}$$

where:

- Y_i is **total wealth**
- D_i is **the treatment indicator for 401(k) participation or eligibility**
- X_i is **the set of household-level controls used in the pension application**

4 Replication Results

The following is the replicated table/figure/results.

The main thing to point out is **the replicated ATE and ATET estimates match the values reported in the original paper**.

Table 1: Replication of ATE and ATET Estimates for 401(k) Plan Participation

Treatment Effect	Estimate	Standard Error
ATE	10180.09	1930.68
ATET	12628.46	2944.43

This result tells us in plain English that **401(k) participation or eligibility is associated with higher total wealth after adjusting for observed covariates using high-dimensional treatment effect methods**. The ATE describes the average effect for the full population, while the ATET describes the average effect for the treated group.

Compared to the original paper, the replicated ATE and ATET estimates are **the same** as the published estimates. Possible reasons for this include **the hdm package includes the same pension dataset and the paper provides directly reproducible R code for the 401(k) application**.

5 Challenges in Replication

The challenges in replicating the results were the following:

- Understanding which section of the paper was narrow enough to replicate
- Distinguishing between ATE, ATET, LATE, and LATET
- Identifying the correct dataset, since the paper contains several empirical applications
- Making sure the R code uses the same simple model specification as the paper

One important lesson from the replication process is that **a paper can be easier to replicate when the authors provide the data, code, and package functions in one place, but the researcher still needs to carefully identify the exact result and model specification being replicated**.

6 Takeaways

The following would be the main takeaways of the paper:

1. **LASSO and Post-LASSO can be used for valid inference in high-dimensional econometric settings**
2. **Orthogonalized treatment effect methods help estimate causal parameters after variable selection**
3. **In the 401(k) application, the estimated ATET is larger than the ATE, meaning the estimated effect is larger for the treated group than for the full sample**

Overall, the paper contributes to the literature by **providing an accessible R package for high-dimensional econometric estimation, inference, instrumental variables, and treatment effects.**

7 Extensions

The following would be possible extensions of the paper:

- **Replicate the LATE and LATET estimates using 401(k) eligibility as an instrument**
- **Compare the simple model to the model with all interaction effects**
- **Compare the hdm treatment effect estimates to simpler OLS or matching-based estimates**

Future work could also explore **how sensitive the 401(k) treatment effect estimates are to alternative control sets, bootstrap standard errors, or different measures of wealth.**

References

- Belloni, A., Chernozhukov, V., and Hansen, C. (2014). Inference on treatment effects after selection among high-dimensional controls. *The Review of Economic Studies*, 81(2):608–650.
- Chernozhukov, V., Hansen, C., and Spindler, M. (2016). High-dimensional metrics in r. *The R Journal*, 8(2):185–199.